

Vitamin B-12 supplementation improves arterial function in vegetarians with subnormal vitamin B-12 status.

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OBJECTIVE: Vegetarians are more vascular-healthy but those with subnormal vitamin B-12 status have impaired arterial endothelial function and increased intima-media thickness. We aimed to study the impact of vitamin B-12 supplementation on these markers, in the vegetarians. **DESIGN:** Double-blind, placebo controlled, randomised crossover study. **SETTING:** Community dwelling vegetarians. **PARTICIPANTS:** Fifty healthy vegetarians (vegetarian diet for at least 6 years) were recruited. **INTERVENTION:** Vitamin B-12 (500 µg/day) or identical placebo were given for 12 weeks with 10 weeks of placebo-washout before crossover (n=43), and then open label vitamin B-12 for additional 24 weeks (n=41). **MEASUREMENT:** Flow-mediated dilation of brachial artery (FMD) and intima-media thickness (IMT) of carotid artery were measured by ultrasound. **RESULTS:** The mean age of the subjects was 45±9 years and 22 (44%) were male. Thirty-five subjects (70%) had serum B-12 levels <150 pmol/l. Vitamin B-12 supplementation significantly increased serum vitamin B-12 levels (p<0.0001) and lowered plasma homocysteine (p<0.05). After vitamin B-12 supplementation but not placebo, significant improvement of brachial FMD (6.3±1.8% to 6.9±1.9%; p<0.0001) and in carotid IMT (0.69±0.09 mm to 0.67±0.09 mm, p<0.05) were found, with further improvement in FMD (to 7.4±1.7%; p<0.0001) and IMT (to 0.65±0.09 mm; p<0.001) after 24 weeks open label vitamin B-12. There were no significant changes in blood pressures or lipid profiles. On multivariate analysis, changes in B-12 (β=0.25; p=0.02) but not homocysteine were related to changes in FMD, (R=0.32; F value=3.19; p=0.028). **CONCLUSIONS:** Vitamin B-12 supplementation improved arterial function in vegetarians with subnormal vitamin B-12 levels, proposing a novel strategy for atherosclerosis prevention.